MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 1AD4

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, subminiature, rf sharp cutoff

Outline --- 2-1 (EIA), metallic shield coating

Base --- Five inline leads, tinned

Envelope --- T2X3

Cathode --- Coated filament

Base connections:

Pin No.	---	1	2	3	4	5
Element	---	a Red dot	g2	-f 1g3 sd	g1	+f 2g3 (Note 1)

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Rg1	Ik	Alt
Unit:	V	Vdc	Vdc	Vdc	Meg	mAdc	ft
Maximum:	1.50	100	---	100	---	7.0	Note 2
Minimum:	1.00	---	---	---	---	---	---
TEST CONDITIONS:	1.25	45	0	45	2	---	---

GENERAL:

Qualification - Required

(F) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection							
(F) 1202	Filament burnout		---	---	---	---	---	---
	Quality conformance inspection, part 1							
1266	Total grid current	Eb = Ec2 = 55 Vdc; Ec1 = -1.0 Vdc; Rg1 = 1 Meg (max) (see note 3)	0.65	II	Ic1	---	-0.5	μAdc
1306	Transconductance (1)	Ef = 1.0 Vdc (see note 3)	0.65	II	ΔSm	1,200	2,500	μmhos
1256	Electrode current (anode)		0.65	II	Ib	1.9	4.1	mAdc
1201	Short and discontinuity detection		0.4	II	----	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Rp = 10,000 ohms	---	---	Ep	---	200	mVac
1016	Moisture resistance	See notes 4 and 5	---	---	---	---	---	---
1301	Filament current		---	---	If	88	112	mA
1256	Electrode current (screen)		---	---	Ic2	0.5	1.3	mAdc
1306	Transconductance (2)		---	---	Sm	1,500	2,500	μmhos
1311	Anode resistance		---	---	rp	0.2	---	Meg
1331	Direct-interelectrode capacitance		6.5	Code E	Cgp Cin Cout	--- 3.0 3.0	0.1 5.0 5.0	pF pF pF
1116	Lead fatigue		4.0	S3	---	---	---	---
1306	Transconductance (3)	Ef = 1.0 Vdc; take reading after 15 minutes	4.0	S3	Sm	1,200	2,500	μmhos
(F) 1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life	Group A: Ef = 1.25 V; Rg1 = 5.0 Meg	---	---	---	---	---	---
---	Intermittent life-test end point (500 hours)	Transconductance (2)	---	---	Sm	1,200	---	μmhos

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NOTES:

1. Grid (3) is composed of two separate deflector plates, one of which is connected to pin 3 and the other to pin 5.
2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in basic document.
3. This test to be performed at the conclusion of the holding period.
4. At the completion of the moisture resistance test, and after a 2-hour holding period, there shall be no evidence of impairment of the coating after rubbing with a soft chamois held firmly in the hand. Exerting as much pressure as possible, the metallic coating shall be rubbed 12 times. Impairment of the coating in the form of blisters, cracks, and peeling shall be cause of rejection. Ten cycles of method 106 of MIL-STD-202 shall be performed; however, step 7 of method 106 shall be omitted.
5. (F) This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 6-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 17
DSA - ES

User activities:
Army - WC
Navy - AS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2516)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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DEPARTMENT OF THE NAVY
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-26254

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